Complex Numbers: Cartesian Form & Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

We are asked to find the difference between two complex numbers, z_1 and z_2. Given $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. We need to calculate $z_1 - z_2$. $z_1 - z_2 = (5 - 2i) - (1 + 4i)$ To subtract complex numbers, we subtract the real parts and the imaginary parts separately. Real part: $5 - 1 = 4$ Imaginary part: $(-2i) - (4i) = -2i - 4i = -6i$ Therefore, $z_1 - z_2 = 4 - 6i$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the complex number w such that $w = x + yi$, where x and y are real numbers. Given that $w^3 = -64i$ and the imaginary part of w is positive, find the value of $x^2 + y^2$.

We are given that $w^3 = -64i$. We can express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2} + 2k\pi$ for integer k. Thus, $-64i = 64e^{i(\frac{3\pi}{2} + 2k\pi)}$. Let $w = re^{i\theta}$. Then $w^3 = r^3e^{i3\theta}$. Equating the moduli, we have $r^3 = 64$, so $r = 4$ (since r must be real and non-negative). Equating the arguments, we have $3\theta = \frac{3\pi}{2} + 2k\pi$. Therefore, $\theta = \frac{\pi}{2} + \frac{2k\pi}{3}$. For $k = 0$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.

The points of intersection A and B are found by setting the two functions equal to each other: $f(x) = g(x)$. We are given $f(x) = x^2 - 4x + 3$ and from the diagram, the line is $y = x - 1$, so $g(x) = x - 1$. Setting them equal: $x^2 - 4x + 3 = x - 1$. Rearranging the equation to form a quadratic equation: $x^2 - 4x - x + 3 + 1 = 0$, which simplifies to $x^2 - 5x + 4 = 0$. We can factor this quadratic equation: $(x - 1)(x - 4) = 0$. The solutions for x are $x = 1$ and $x = 4$. We are given that point A has an x-c.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of b.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(1, 4)$, we have $f(x) = a(x - 1)^2 + 4$. The parabola passes through the point $(-1, 0)$. Substituting these coordinates into the equation: $0 = a(-1 - 1)^2 + 4$ $0 = a(-2)^2 + 4$ $0 = 4a + 4$ $4a = -4$ $a = -1$. So, the function is $f(x) = -1(x - 1)^2 + 4$. Expanding this: $f(x) = -(x^2 - 2x + 1) + 4$ $f(x) = -x^2 + 2x - 1 + 4$ $f(x) = -x^2 + 2x + 3$. Comparing this to the general form.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

(a) Vertical Asymptote: The denominator is zero at $x = 3$. Numerator at $x = 3$ is $3^2 - 3 - 2 = 4eq0$. Thus $x = 3$ is a vertical asymptote. Oblique Asymptote: Perform polynomial long division: $\frac{x^2 - x - 2}{x - 3} = x + 2 + \frac{4}{x - 3}$. As $x \to \pm\infty$, $\frac{4}{x - 3} \to 0$. Therefore, $y = x + 2$ is the oblique asymptote. (b) Differentiate $f(x) = x + 2 + 4(x - 3)^{-1}$: $f'(x) = 1 - \frac{4}{(x - 3)^2}$. Set $f'(x) = 0$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Show that the sum of these five roots equals 0.

1. Write $-32 = 32e^{i\pi}$. By De Moivre's Theorem: $z_k = 32^{1/5}\exp\left(i\frac{\pi + 2\pi k}{5}\right) = 2e^{i(\pi + 2\pi k)/5}$ for $k = 0, 1, 2, 3, 4$. Evaluating the arguments in $(-\pi, \pi]$: - $k = 0$: $z_0 = 2e^{i\pi/5}$ - $k = 1$: $z_1 = 2e^{i3\pi/5}$ - $k = 2$: $z_2 = 2e^{i5\pi/5} = 2e^{i\pi} = -2$ - $k = 3$: $z_3 = 2e^{i7\pi/5} = 2e^{-i3\pi/5}$ - $k = 4$: $z_4 = 2e^{i9\pi/5} = 2e^{-i\pi/5}$ 2. Sum of the roots: Notice that.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.

Since triangles ABC and ADE are similar, the ratio of corresponding sides is equal. The ratio of similarity between triangle ADE and triangle ABC can be found by comparing the lengths of corresponding sides AD and AB. We have AB = AD + DB = 6 + 4 = 10 cm. The ratio of similarity, k, is $\frac{AB}{AD} = \frac{10}{6} = \frac{5}{3}$. Alternatively, we can consider the ratio of similarity from ADE to ABC as $\frac{AD}{AB} = \frac{6}{10} = \frac{3}{5}$. Using this ratio, we can find the length of BC. The corresponding s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

1. Normal vectors: $\mathbf{n}_1 = \langle 1, 1, 1 \rangle$, $\mathbf{n}_2 = \langle 2, -1, 3 \rangle$. 2. Direction vector $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$: $\mathbf{d} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 1, 1; 2, -1, 3| = \mathbf{i}(3 - (-1)) - \mathbf{j}(3 - 2) + \mathbf{k}(-1 - 2) = (4; -1; -3)$. 3. Find a point on the line: set $z = 0$: $\{x + y = 1; 2x - y = 4 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3},\ y = 1 - \frac{5}{3} = -\frac{2}{3}$. P.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the complex number z such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that a is a real number, find the possible values of a.

The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. We are given $z = a + 5i$ and $|z| = \sqrt{26}$. Therefore, $|z| = \sqrt{a^2 + 5^2} = \sqrt{a^2 + 25}$. We are given that $|z| = \sqrt{26}$. So, we have the equation: $\sqrt{a^2 + 25} = \sqrt{26}$. Squaring both sides gives: $a^2 + 25 = 26$. Subtracting 25 from both sides, we get: $a^2 = 1$. Taking the square root of both sides, we find: $a = \pm 1$. Common mistakes include: ignoring the fact that a can be negative, or incorrectly.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let $A = (x_1, y_1, z_1) = (1, 2, 3)$ and $B = (x_2, y_2, z_2) = (4, 6, 9)$. The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1; z_2 - z_1)$. Substituting the given coordinates: $\vec{AB} = (4 - 1; 6 - 2; 9 - 3) = (3; 4; 6)$. Therefore, the vector $\vec{AB}$ in component form is $(3; 4; 6)$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.

The function is given by $f(x) = \frac{x^2 - ax + 6}{x - 1}$. $x = 1$ is a vertical asymptote, which means that $x - 1$ is not a factor of the numerator, and when $x = 1$, the numerator is non-zero. The point of discontinuity is at $(2, k)$. This implies that $x = 2$ is a root of the numerator, so $f(2) = 0$ or the expression is undefined at $x = 2$ if $x - 2$ were a factor of the denominator. Since the denominator is $x - 1$, which is not zero at $x = 2$, the function must be zero at $x = 2$ for there to be a removable discontinuity (a.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of k. The graph of $y = f(x)$ is shown below.

For the function $f(x) = \frac{x^2 - 9}{x - k}$ to have a removable discontinuity at $x = 3$, the numerator and the denominator must both be zero at $x = 3$. This means that $(x - 3)$ must be a factor of both the numerator and the denominator. The numerator is $x^2 - 9$, which factors as $(x - 3)(x + 3)$. For the denominator to have a factor of $(x - 3)$, we must have $x - k = 0$ when $x = 3$. Thus, $3 - k = 0$, which implies $k = 3$. When $k = 3$, $f(x) = \frac{(x - 3)(x + 3)}{x - 3}$. For $xeq3$, $f(x) = x + 3$. The discontinuit.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.

1. Normalization condition: $\int_0^2 k(2x - x^2)\,dx = 1 \Rightarrow k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left(4 - \frac{8}{3}\right) = k\left(\frac{4}{3}\right) = 1 \Rightarrow k = \frac{3}{4}$. 2. Median m: By visual symmetry of the parabola $f(x) = \frac{3}{4}x(2 - x)$ about $x = 1$, $m = 1$. We verify: $F(1) = \int_0^1 \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{2}{3}\right) = \frac{1}{2}$. ✓ 3. Variance: By symmetry, $E[X] = 1$..

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.

To find the local minimum, we first need to find the derivative of the function $f(x)$. \\ \\ $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. \\ \\ Local extrema occur where $f'(x) = 0$. So, we set $3x^2 - 12x = 0$. \\ \\ Factoring out $3x$, we get $3x(x - 4) = 0$. \\ \\ This gives us two critical points: $x = 0$ and $x = 4$. \\ \\ To determine whether these points are local maxima or minima, we can use the second derivative test. \\ \\ The second derivative is $f''(x) = \frac{d}{dx}(3x^2 - 12x) = 6x - 12$. \\.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?

To find the x-intercepts of the function $f(x) = x^2 - 6x + 8$, we need to solve for x when $f(x) = 0$. This means we need to find the roots of the quadratic equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x - 2)(x - 4) = 0$. Setting each factor to zero gives us the x-intercepts: $x - 2 = 0 \Rightarrow x = 2$ and $x - 4 = 0 \Rightarrow x = 4$. The graph sho.

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16 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2 \cos x\,dx = [x^2 \sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x \sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x \sin x\,dx = \frac{\pi^2}{4} - 2J.$ 2. Second IBP on $J = \int_0^{\pi/2} x \sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Proof by Mathematical Induction**

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2 + y^2 = r^2$, $x = r\cos\theta$, and $y = r\sin\theta$: $(r^2)^2 = 4(r^2\cos^2\theta - r^2\sin^2\theta) \Rightarrow r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$. Apply the double-angle identity $\cos^2\theta - \sin^2\theta = \cos(2\theta)$: $r^4 = 4r^2\cos(2\theta)$. Divide through by r^2 (for $req0$): $r^2 = 4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r = 2$ at.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

1. Step formula: $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h(x_n + y_n)$ with $h = 0.1$, $x_0 = 0$, $y_0 = 1$. Step 1 (to $x_1 = 0.1$): $y_1 = y_0 + 0.1(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.1$. Step 2 (to $x_2 = 0.2$): $y_2 = y_1 + 0.1(x_1 + y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22$. Euler's approximation is $y(0.2) \approx 1.22$. 2. Exact analytical solution: Rewrite as $\frac{dy}{dx} - y = x$. Integrating factor $\mu = e^{-x}$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.04 Partial Fractions**

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

1. Compute the cross product via determinant: $\mathbf{a} \times \mathbf{b} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2, -1, 3; 1, 4, -2| = \mathbf{i}[(-1)(-2) - 3(4)] - \mathbf{j}[2(-2) - 3(1)] + \mathbf{k}[2(4) - (-1)(1)]$ $= \mathbf{i}(2 - 12) - \mathbf{j}(-4 - 3) + \mathbf{k}(8 + 1) = -10\mathbf{i} + 7\mathbf{j} + 9\mathbf{k} = (-10; 7; 9)$. 2. The area of the triangle is half the magnitude of the cross product.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$.
$\cot(2\theta) = \frac{A-C}{B} = \frac{5-8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$.
Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1+\cos(2\theta)}{2}} = \sqrt{\frac{1+3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$, .

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

a. $5 + 14i$ ✓

b. $-3 - 8i$

c. $5 - 14i$

d. $-3 + 14i$

$z_1 z_2 = (3 - 2i)(-1 + 4i) = -3 + 12i + 2i - 8i^2 = -3 + 14i + 8 = 5 + 14i$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $\frac{4+2i}{1-i}$ in Cartesian form $a + bi$.

a. $1 + 3i$ ✓

b. $3 + 3i$

c. $1 - 3i$

d. $2 + 2i$

Multiply numerator and denominator by conjugate $(1 + i)$: $\frac{(4+2i)(1+i)}{(1-i)(1+i)} = \frac{4+4i+2i+2i^2}{1-i^2} = \frac{2+6i}{2} = 1 + 3i$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = x + iy$, what is $z \cdot z^*$ (where z^* is the complex conjugate)?

a. $x^2 + y^2$ ✓

b. $x^2 - y^2$

c. $x^2 + 2ixy - y^2$

d. $2x$

$zz^* = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 - i^2y^2 = x^2 + y^2 = |z|^2$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

a. $|z| = 2$, $\text{Arg}(z) = \frac{5\pi}{6}$ ✓

b. $|z| = 2$, $\text{Arg}(z) = -\frac{\pi}{6}$

c. $|z| = 4$, $\text{Arg}(z) = \frac{2\pi}{3}$

d. $|z| = 2$, $\text{Arg}(z) = \frac{7\pi}{6}$

$|z| = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$. In Quadrant II ($x < 0, y > 0$), reference angle is $\arctan(1/\sqrt{3}) = \frac{\pi}{6}$. Thus $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus of $\frac{3+4i}{1-2i}$?

a. $\sqrt{5}$ ✓

b. $\frac{5}{3}$

c. 5

d. $\frac{25}{5} = 5$

Using modulus properties: $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} = \frac{\sqrt{3^2+4^2}}{\sqrt{1^2+(-2)^2}} = \frac{5}{\sqrt{5}} = \sqrt{5}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the principal argument of $z = -5i$.

a. $-\frac{\pi}{2}$ ✓

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{2}$

d. π

The point lies on the negative imaginary axis, which corresponds to angle $-\frac{\pi}{2}$ within the principal range $(-\pi, \pi]$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the two square roots of $-8 - 6i$ in Cartesian form.

a. $\pm(1 - 3i)$ ✓

b. $\pm(3 - i)$

c. $\pm(1 + 3i)$

d. $\pm(2 - 3i)$

Let $(x + iy)^2 = x^2 - y^2 + 2ixy = -8 - 6i$. Thus $x^2 - y^2 = -8$ and $2xy = -6 \Rightarrow y = -3/x$.
$x^2 - 9/x^2 = -8 \Rightarrow x^4 + 8x^2 - 9 = 0 \Rightarrow (x^2+9)(x^2-1) = 0$. Since $x \in \mathbb{R}$, $x = \pm 1$, giving $y = \mp 3$. Roots are $\pm(1 - 3i)$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the quadratic equation $z^2 - 4z + 13 = 0$ over $\mathbb{C}$.

a. $z = 2 \pm 3i$ ✓

b. $z = 4 \pm 6i$

c. $z = -2 \pm 3i$

d. $z = 2 \pm 9i$

By quadratic formula: $z = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Describe the locus of points in the complex plane satisfying $|z - 3 + 2i| = 5$.

a. **A circle with center** $(3, -2)$ **and radius** 5 ✓

b. A circle with center $(-3, 2)$ and radius 5

c. A straight line perpendicular to $3 - 2i$

d. An ellipse with focal distance 5

Rewrite as $|z - (3 - 2i)| = 5$. The distance from z to the fixed point $3 - 2i$ is constant and equal to 5, which defines a circle centered at $(3, -2)$ with radius 5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What geometric locus is defined by $|z - 1| = |z - 3i|$?

a. **The perpendicular bisector of the segment connecting** $(1, 0)$ **and** $(0, 3)$ ✓

b. A circle passing through $(1, 0)$ and $(0, 3)$

c. A line segment connecting $(1, 0)$ and $(0, 3)$

d. A parabola opening upward

$|z - z_1| = |z - z_2|$ represents the set of all points equidistant from $z_1 = 1 + 0i$ and $z_2 = 0 + 3i$, which is the perpendicular bisector of the segment connecting them.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the Cartesian equation of the locus $\text{Arg}(z-2)=\frac{\pi}{4}$.

a. **The half-line $y=x-2$ for $x>2$** ✓

b. The full line $y=x-2$

c. The circle $(x-2)^2+y^2=1$

d. The half-line $y=-x+2$ for $x<2$

$\text{Arg}((x-2)+iy)=\frac{\pi}{4}\Rightarrow\frac{y}{x-2}=\tan(\pi/4)=1\Rightarrow y=x-2$, with the ray restriction $x-2>0\Rightarrow x>2, y>0$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct complex nth roots does the equation $z^7=1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 7 roots, $\frac{360}{7}^\circ$ apart ✓

b. 8 roots, $\frac{360}{7}^\circ$ apart

c. 7 roots, 45° apart

d. 6 roots, $\frac{360}{7}^\circ$ apart

$z^7=1$ has exactly 7 roots, equally spaced by $\frac{360^\circ}{7}=\frac{360}{7}^\circ$.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $z=x+yi$ be a complex number. The modulus of z, denoted by $|z|$, is given by $|z|=\sqrt{x^2+y^2}$. Given that $|z|=\sqrt{5}$ and the real part of z is 2, find the possible values for the imaginary part of z.

a. $\pm\sqrt{3}$ ✓

b. ±3

c. $\pm\sqrt{1}$

d. ±1

We are given that $z=x+yi$. The modulus of z is $|z|=\sqrt{x^2+y^2}$. We are given that $|z|=\sqrt{5}$, so $\sqrt{x^2+y^2}=\sqrt{5}$. Squaring both sides, we get $x^2+y^2=5$. We are also given that the real part of z is 2, which means $x=2$. Substituting $x=2$ into the equation $x^2+y^2=5$, we get $2^2+y^2=5$, which simplifies to $4+y^2=5$. Subtracting 4 from both sides, we have $y^2=1$. Taking the square root of both sides, we find $y=\pm1$. Therefore, the possible valu.

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. **5** ✓

b. 7

c. 25

d. $\sqrt{7}$

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the given complex number $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus of z is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Common misconceptions: - Adding the real and imaginary parts: $3 + 4 = 7$, which is incorrect. - Squaring the real and imaginary parts without taking the square root: $3^2 + 4^2 = 9 + 16 = 25$, which is the square of the modulus. - Incorrectly calculating the square root.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = 3 + ki$, where k is a real parameter. The modulus of z is given by $|z| = \sqrt{13}$. Find the positive value of k.

a. **1** ✓

b. 2

c. 3

d. 4

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the complex number $z = 3 + ki$, we have $a = 3$ and $b = k$. Therefore, $|z| = \sqrt{3^2 + k^2} = \sqrt{9 + k^2}$. We are given that $|z| = \sqrt{13}$. So, we have the equation: $\sqrt{9 + k^2} = \sqrt{13}$ Squaring both sides of the equation, we get: $9 + k^2 = 13$ Subtracting 9 from both sides: $k^2 = 13 - 9$ $k^2 = 4$ Taking the square root of both sides, we get $k = \pm 2$. Since we are asked for the positive value of k, w.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = -3 + 4i$. Find the modulus of z, denoted by $|z|$.

a. **5** ✓

b. 7

c. 25

d. 16

The modulus of a complex number $z = a + bi$ is given by the formula $|z| = \sqrt{a^2 + b^2}$. In this case, $a = -3$ and $b = 4$. Therefore, $|z| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

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37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A cubic equation with real coefficients $x^3 - 10x^2 + 37x - 52 = 0$ has $3 + 2i$ as one root. Find the real root.

a. 2

b. **4 ✓**

c. 3

d. 0

Complex roots occur in conjugate pairs, so $3 - 2i$ is also a root. Product of all three roots $= -d = 52$, and $(3 + 2i)(3 - 2i) = 13$, so the real root $= \frac{52}{13} = 4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

For which value of m does the system $x + my = 3,\ 2x + 4y = 6$ have infinitely many solutions?

a. **2 ✓**

b. 4

c. 1

d. 3

For infinitely many solutions the second equation must be exactly twice the first: $2/1 = 2$ and $4/m = 2 \Rightarrow m = 2$ (and $6 = 2 \times 3$, consistent).

39 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.04 Partial Fractions**

How many solutions does the system $x + y = 4,\ 2x + 2y = 8$ have?

a. **Infinitely many ✓**

b. No solution

c. Exactly one

d. Exactly two

The second equation is exactly twice the first, so both describe the same line — infinitely many solutions.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find all cube roots of unity.

a. $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$ ✓

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

$z^3 = 1 \Rightarrow z_k = e^{2\pi ik/3}$ for $k = 0, 1, 2$: $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

a. $|z^6| = 64,\ \arg(z^6) = 60^\circ$

b. $|z^6| = 12,\ \arg(z^6) = 0^\circ$

c. $|z^6| = 64,\ \arg(z^6) = 90^\circ$

d. $|z^6| = 64,\ \arg(z^6) = 0^\circ$ ✓

By De Moivre's theorem, $|z^6| = |z|^6 = 2^6 = 64,\ \arg(z^6) = 6 \times 60^\circ = 0^\circ$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

In the expansion of $(1 + x)^7$, find the ratio of the coefficient of x^4 to the coefficient of x^3.

a. 1 ✓

b. 2

c. 3

d. 6

Working through this complex-numbers step: $\frac{\binom{7}{4}}{\binom{7}{3}} = \frac{7-3}{3+1} = 1$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find $\bar{z}$ (the complex conjugate) if $z = -3 + 9i$.

a. $-3 - 9i$ ✓

b. $-3 + 9i$

c. $3 - 9i$

d. $3 + 9i$

The conjugate negates the imaginary part: $\bar{z} = -3 - 9i$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the coefficient of x^5 in the expansion of $(1 + 2x)^7$.

a. 519

b. 780

c. 787

d. **672** ✓

Working through this complex-numbers step: $751^2 2^5 = 21132 = 672$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Which value of k makes the system $x + 2y = 5,\ 2x + 4y = k$ have infinitely many solutions?

a. **10** ✓

b. 5

c. 20

d. 15

The second equation must be exactly twice the first: $k = 2(5) = 10$.

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